

Amar Dhillon

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EDUCATION

University of Maryland, College Park
B.S. in Computer Science, Minor in Statistics; GPA: 3.5

College Park, MD
Expected May 2028

CERTIFICATIONS

Amazon Web Services (AWS) Certified Developer - Associate
Amazon Web Services

Nov 2024 – Nov 2027
Digital Badge

PROJECTS

IEX Order Book & Matching Engine | *C++17, STL, POSIX, Linux perf* Jul 2026 – Aug 2026

- Built a C++17 order book and matching engine to replay **50.9M** real IEX exchange messages without ever resyncing, which agreed with the exchange on 99,122 of 99,123 fills and 100% of top-of-book prices
- Designed the order book with std::list FIFOs per price level in a std::map and an order-ID hash map to make cancels $O(1)$, which ran the full replay at **1.26M events/sec** and hit a 1.2 μ s p99 on a separate 2M-event run
- Implemented B+ tree and LSM-tree backends for the engine's event log to compare them on identical 2M-record workloads, which showed the B+ tree giving **12.8x** the read throughput and 22x faster range queries for 27% more disk
- Traced an apparent 1.65x regression to the CPU frequency governor with perf counters and a do-nothing control book to rule out a code change, which led to core pinning that cut run-to-run variance from 6% to 0.06%

EXPERIENCE

Software Development Engineer Intern
Amazon Web Services

Sep 2026 – Dec 2026
Redmond, WA

- Writing backtested Java capacity-planning logic that sets upper and lower bounds on the EC2 capacity Amazon Redshift reserves for patching to balance availability against cost

Software Engineering Intern
Capital One

Jun 2026 – Aug 2026
Richmond, VA

- Cut AI workflow generation time **65%** to under 15s by splitting a Claude skill into focused sub-skills that call Python search and validation tools instead of relying on open-ended model search
- Built the Claude skill's main search tool by precomputing popular workflows from a 270K+ row PostgreSQL table into a Redis-cached category tree to keep lookups within the database's permission and rate limits

Software Engineer (Contract)
UMD App Development Contracting (Client: Amazon Leo)

Jan 2026 – Present
College Park, MD

- Reduced end-to-end latency **34%** for a pipeline routing 6,000+ scraped research documents to inference by replacing S3 staging hops with Kinesis streams feeding Lambda

Machine Learning Research Intern
UMD Institute for Advanced Computer Studies (UMIACS)

Dec 2025 – Present
College Park, MD

- Built and prompt-tuned a pipeline for a PhD-led robotics project aimed at ICLR that overlays AnyGrasp's candidate grasps on camera images in Open3D to help a vision-language model pick grasps that avoid hitting the table

Co-Founder, Software Engineer
TerpLabs

Jun 2025 – Present
College Park, MD

- Built Redis caching in front of Tortuga's Prisma ORM queries to reduce database load and latency for a class scheduler that drew 20,000+ visits in its first 30 days

TECHNICAL SKILLS

Languages: C++17, C, Python, Java, Rust, SQL, TypeScript, Bash

Systems & Performance: debugging and profiling with Linux perf (hardware counters, top-down analysis), POSIX, STL, B+ trees, LSM trees, data structures and algorithms, benchmarking

Cloud & Tools: AWS (Lambda, Kinesis, EC2, Fargate, CloudWatch monitoring), AWS CDK, PostgreSQL, Redis, Docker, GitHub Actions (CI/CD), Git, unit testing (Pytest)